

Appendix A for SC27: Training and Portability — What MoA-Derived Kernels Have Actually Demonstrated

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\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{graphicx}
\usepackage{amsmath}
\usepackage{hyperref}
\usepackage{booktabs}
\usepackage{listings}
\lstset{basicstyle=\ttfamily\footnotesize,breaklines=true}

\title{Appendix A for SC27: Training and Portability — What MoA-Derived Kernels Have Actually Demonstrated}
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\begin{document}
\maketitle

\begin{abstract}
This appendix distills extended real-hardware validation into essential findings for SC27. Across two NVIDIA GPU ar
\end{abstract}

\section{Universal IR Principle (Row 2)}
Any language after syntactic sugar removed has AST with shape and components. Walk and optimize by combining indexe

Then DNF → offset:  $\text{Ivec } \psi A \text{ lequiv } (\text{rav } A) \backslash \gamma (\text{ivec}, \text{rho } A) \backslash \psi A \text{ lequiv } A$ . Family of

$$\sum_k \sum_{j>k} s_j \cdot \gamma^{\text{col}}(v, s) = \sum_k v_k \prod_{j<k} s_j \cdot \gamma^{\text{Numa}}(v$$


Dimension lifting:  $(\prod \text{rho } A)_{\text{data}}$  must be  $(\prod \text{rho } A)_{\text{machines}}$  where  $i \in \text{kin } j \in \text{langl}$ 

This is what scientist knows — math once, today/yesterday/tomorrow portable via standardized tools providing determ

\section{What Was Tested (Papers I-III)}
\begin{itemize}
\item Forward-pass attention (Paper I) — fused, memory-optimal online-softmax Triton, fp32/fp16/fp64
\item Decode attention (Paper III) — single-query KV-cache-resident,  $K_{\text{new}} = K_{\text{old}} \cdot \frac{1}{k} \cdot \frac{1}{t} \cdot \frac{1}{O(dk+dv)}$  via cat
\item Fused backward pass (Paper II) —  $G_A, G_S, G_{\text{sc}}$  as scalars:  $G_{\text{sc}} = \langle \psi d \rangle \cdot \langle k \rangle \cdot \psi V, \text{sdS}_{\text{ik}} = Y_{\text{ij}}$ 
\end{itemize}
Ported A100→H100 with no code changes, plus OpenACC via nvc.

\section{Finding 1: Memory-Optimality Has No Cliff (3 Demonstrations)}
\begin{itemize}
\item Precision (forward) at fp64  $B=4, n=8192$  PyTorch SDPA fallback OOMs 16 GiB explicit  $(B, H, n, n)$  matr
\item Scale across GPUs (backward)  $B=4, n=8192$  fp32 unfused needs several  $(B, H, n, n)$  tensors, exhausts 4
\item Scale on same GPU MoA unfused training-step at  $B=4, n=8192$  exhausts memory, same inputs same hardwa
\end{itemize}

\section{Finding 2: Portability — Correctness Transfers, Performance Improves}
Every kernel ported A100→H100 no code changes:
\begin{itemize}
\item Correctness transfers without re-verification: all fp32/fp16/fp64 passed at H100 at  $\sim 5.960 \times 10^{-8}$ 
\item Performance does not merely transfer — it improves in every case: Decode  $7.2 \times \rightarrow 13.9 \times$  over PyTorch
\item Third realization OpenACC kernels also ported to H100 with same  $\text{num\_workers}(16)$ ,  $\text{vector\_length}(32)$  reused
\item Third GPU generation V100 (SDSC Expanse) — 15 correctness checks passed fp64, extends claim. Performance anom
\end{itemize}

\section{Finding 3: Deriving Kernel Structure From Machine Shape Works}
Same fused backward algorithm realized second way OpenACC via NVIDIA HPC SDK nvc:
\begin{itemize}
\item Literal scalar translation correct but 6-70% slower than every baseline including naive unfused Python
\item Restructured to expose explicit  $\text{sgang}/\text{worker}/\text{vector}$  parallelism model onto A100's actual known hardware shap
\end{itemize}
\begin{itemize}
\item  $\text{sgang}$  to one  $(b, ir)$  pair — same axis as Triton grid
\item  $\text{worker}$  with  $\text{num\_workers}(16)$  to key/value-position loop distributed across A100's 16 warps per scheduler
\item  $\text{vector}$  with  $\text{vector\_length}(32)$  to head-dim reduction loops, matched to 32-wide warp
\end{itemize}

Results A100  $B=2$  real hardware:
\begin{table}[h]\centering
\begin{tabular}{rrrr}
\toprule  $n$  &  $v1$  (ms) &  $v2$  (ms) & speedup \\ \midrule 512 & 32.3 & 2.0 & 16.5 \\ 2048 & 132.7 & 24.8 & 5.4
\end{tabular}
\end{table}

\begin{table}[h]\centering
\begin{tabular}{rrrr}
\toprule  $n$  & vs SDPA & vs unfused & vs Triton fused \\ \midrule 512 & 4.3 & 2.7 & 1.4 \\ 20
\end{tabular}
\end{table}
From  $6-23 \times$  to  $1.4-2.0 \times$  of hand-tuned Triton via compiler-mediated scheduling given explicit parallel

Forward-only OpenACC did NOT repeat this on A100/H100: 6-96% slower than Triton/SDPA with disproportionate s

Measuring actual hardware shape via \nvaccelinfo (NVIDIA introspection): warp size, max threads/SM, shared

\section{Standardized Compiler Flags and Epsilon — Predictive Theory Accuracy}
Universal IR requires deterministic tools providing time, power, energy, heat:

\begin{lstlisting}
// C OpenMP deterministic for T~alpha F+beta M
-O2 -fno-fast-math -fopenmp -DOMP_DETERMINISTIC -fopenmp-targets=nvptx64
#pragma omp parallel for schedule(static) proc_bind(close)

// Fortran OpenACC deterministic + shape disclosure
```

```
-acc -Minfo=accel,mem -ta=tesla:cc80,cc90 -fopenacc -O2 -fno-fast-math
!$acc parallel loop gang worker vector length(64) independent
// Minfo: gang=grid, worker=num_workers, vector=vector_length, gamma choice
```

```
// OpenMPI weighted comm-free: rho_machine = <numa_nodes # cores>
MPI_Scatterv/Gatherv + hwloc binding, X_d=0% via Level Zero Sysman 99.88% busy
Nsight Compute atomic throughput, stall_memory
\end{lstlisting}
```

$\$-O2$ not $\$-O3$ heuristic, $\$-fno-fast-math$ guarantees $\|\text{err}\|_{\infty} \sim 10^{-15}$ $\$-10^{-16}$ exact across A100/H100

Remaining noise is epsilon — small, measurable, shrinkable as flags improve:
Epsilon examples: V100 clean vs A100/H100 4K to 8K anomaly (L2 capacity), H100 2.0-2.5 $\times$ faster no code chan

```
\texttt{T\_local=524288.0, T\_remote=280494080.0 ratio=535x NUMA}
\texttt{atomics naive 67108864 vs related 33554432 ratio=2.00x -> 2.5x fix via private acc}
\texttt{gamma\_row 6 vs gamma\_col 7, time gap 3.17x matches stall gap 3.35x}
```

All standardized, all shapes.

\section{SC27 Connection — Two IPDPS Submissions, One Theory}

At UA: signal processing combinatorial (RGB $\rightarrow$ Grey, Sobel $P_d=9$ window 3×3) via $\psi \rightarrow \text{DNF} \rightarrow \gamma \rightarrow \text{Ope}$

At ACCESS: AI Attention suite 1-4 (Paper I forward, II backward scalar G_{sc} , III decode $K_{\text{new}}=K_{\text{old}} \setminus k_{\text{t}}$)

Both independent discovery paths yield same predictive variables $T \approx \alpha F + \beta M + \gamma A + \delta C$, $\$E, P$

This appendix provides concrete next experiments (from synthesis Sec 6):

```
\begin{enumerate}
\item Close OpenACC gap to Triton, specifically toward tensor cores — current 1.4-2.0 $\times$  likely tensor-core ga
\item Isolate compiler version vs L2-capacity via controlled repeat: same GPU two nvc releases, same release two GP
\item Derive rather than confirm machine-shape parameter for kernel not yet built — stronger test than confirming k
\item Extend cross-architecture test to third vendor AMD/Intel — test whether portability and shape-derivation clai
\end{enumerate}
```

\section{Artifacts for Reproduction}

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\begin{itemize}
\item \texttt{moa\_compiler/paper1\_exact\_reproduce.py, paper2\_backward.py, paper3\_decode.py, paper45\_validatio}
\item \texttt{moa\_complete\_final.tex, moa\_sc\_demo.ipynb} with validation run \texttt{tmp/run.txt}
\item Flags: \texttt{-O2 -fno-fast-math -fopenmp -DOMP\_DETERMINISTIC} + \texttt{nvaccelinfo} for hardware shape
\item Devices: NCSA Delta A100, PSC Bridges-2 H100, SDSC Expanse V100, AMD MI100, TACC Stampede3 Intel Max 1550
\end{itemize}
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